

A capability boundary for covariance-driven single-perturbation models: the perturbed gene is detected through its own coordinate, not predicted by covariance

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Author summary

A popular class of perturbation-prediction methods assumes that the natural variability of unperturbed cells encodes how those cells will respond to a genetic perturbation, so that a gene–gene covariance matrix can predict the effect of knocking a gene down. These methods report high accuracy, and it is widely assumed they have learned something specific to each perturbed gene. We reproduced the leading such framework (CIPHER; $\Delta X = \Sigma \cdot u$) with the authors’ own code across sixteen large single-cell perturbation datasets and simple baselines they did not report. Its accuracy comes almost entirely from a single shared axis — a technical cell-size/sequencing-depth response — rather than from anything specific to the perturbed gene: a covariance column from a random gene predicts the response as well as the column of the gene actually perturbed, and a single global mode predicts best of all. Yet a real, gene-specific response structure does exist beneath that axis: it is reproducible across laboratories, cell lines, guide libraries and stimulation times, organised by protein–protein interactions and pathways, and visible in the mean, variance, covariance and distribution of responses. The covariance operator simply cannot see it. The same model can therefore detect which gene was perturbed, but detection is carried by the perturbed gene’s own transcript, not by the covariance structure, and the model is not a **predictor** of that gene’s transcriptome-wide response. We distil these results into a reusable specificity-audit protocol, and we show that swapping the data space or the covariance estimator does not help — the limitation is conceptual, not a matter of preprocessing.

Abstract

Predicting transcriptome-wide perturbation responses from the unperturbed state is a central goal of computational biology. The CIPHER framework derives predictions from the gene–gene covariance of control cells ($\Delta X = \Sigma \cdot u$) and reports high accuracy. Reproducing CIPHER with the authors’ released code across sixteen single-cell perturbation datasets (>2.0 million cells; nested subsets of 13/10/8 for the global-mode, predictive and expression-space analyses), we characterise the **capability boundary** of covariance-driven single-perturbation models. We find that (i) the leading eigenvector of the control covariance accounts for 42–87% of its variance, its loadings are almost identical to the gene-mean vector ($|r| = 0.99\text{--}1.00$), and equal-depth resampling halves its share ($0.68 \rightarrow 0.26$); (ii) covariance columns are almost perfectly collinear (median cosine 0.987), a column from a random gene matched on mean/variance/detection equals the perturbed

gene’s column (scale-only 0.437 vs 0.423, $p = 0.49$), and the global mode is the strongest predictor (raw uncentered R^2 0.266 vs full 0.215; full – global = -0.031 [$-0.049, -0.015$]); (iii) forward predictions are almost identical across perturbations (mean pairwise $|\cos|$ 0.93 vs true 0.46) and preserve no perturbation-specific structure (RSA ≈ 0 , $p = 0.32$), whereas the inverse recovers the perturbed gene accurately (AUC 0.62–0.99) but only because of the perturbed gene’s own coordinate (removing it collapses AUC $0.93 \rightarrow 0.37$; removing the covariance changes nothing, $p = 0.74$). Nevertheless, below the depth axis there is a real, reproducible perturbation-specific structure: after removing the global axis and the target’s own coordinate, the split-half reliability of the response-similarity matrix is 0.49 [0.31, 0.67] ($n = 12$ datasets, $p = 5 \times 10^{-4}$); it is broad-spectrum, transfers across laboratories/cell lines/libraries (cosine 0.38–0.49, null ≈ 0) and across stimulation time and donors (0.28–0.48), and is organised by PPI/pathways/TF regulons (PPI edge – non-edge +0.098, $p = 0.023$). The structure is captured by shared low-dimensional programmes (~ 20) whose **amplitudes** are highly reliable (0.85–0.96) plus a **small target-specific direction** (reliability 0.20–0.52 after removing the programmes); individual guides are noisy reporters, and averaging guides recovers the target-level signal (multi-guide cross-dataset transfer 0.14–0.21 $\rightarrow$ 0.35–0.38). Crucially, the covariance/precision operator is blind to this structure: across raw/CPM/log1p/CLR/NB-Pearson spaces and covariance, precision, shrunk, MP-clipped, factor and rank-1 estimators, off-axis predictability stays at chance (pooled -0.004 [$-0.012, 0.003$], $p = 0.52$). Detection of the perturbed gene is therefore carried by that gene’s **own coordinate**, not by the covariance structure, and the same matrix does not predict the response programme; recovering the target-specific direction requires cross-dataset alignment or network priors, not a better covariance estimator. We provide a specificity-audit protocol and recommend matched-random-column, equal-depth and global-mode baselines for all such models.

Introduction

How cells respond to genetic perturbations is central to functional genomics and to the emerging vision of an “AI virtual cell” [1]. Pooled single-cell perturbation screens — Perturb-seq and related methods — now profile the transcriptomic consequences of hundreds to thousands of perturbations [2–5], motivating computational models that predict responses to perturbations never performed experimentally [6–9].

A particularly attractive class of models derives predictions from the **unperturbed** state alone. The CIPHER framework [10] invokes linear-response theory and proposes that the response to a perturbation vector u is $\Delta X = \Sigma \cdot u$, where Σ is the gene–gene covariance of control cells. CIPHER reported strong recapitulation of single- and double-perturbation responses and argued that removing gene–gene covariances reduced performance roughly eleven-fold. In parallel, deep-learning perturbation predictors have been shown not to outperform simple baselines [11]. This raises a question not yet systematically addressed for covariance-based methods: **is the prediction specific to the perturbed gene at all, and if not, is that because no specific signal exists?**

Here we answer both halves of the question for the **single-perturbation forward model** across sixteen datasets, using the authors’ released code together with matched random-column base-

lines, equal-depth resampling, spectral nulls, split-half reproducibility, cross-laboratory/cell-line/library/time comparisons, network and pathway organisation tests, an explicit programme/amplitude/direction decomposition, a transform $\times$ estimator scan, and a small identifiability analysis. Our central result is a **capability boundary**: the same control covariance matrix is sufficient to *identify* the perturbed gene (a detection problem, carried by the gene’s own coordinate) but not to *predict* its gene-specific response (a prediction problem); the target-specific structure that does exist is real and reproducible but lies outside the range of the covariance operator, and must be recovered by other means.

Results

Datasets and analysis subsets

Sixteen single-cell perturbation datasets from the scPerturb collection [12] were analysed (2,044,655 cells; per-dataset counts and inclusion flags in Table S1). Six further datasets downloaded during the exploratory phase (CRISPRa screens, a genome-wide Replogle screen, the XAtlas HCT116/HEK293T panels and a “proper” Perturb-seq dataset) extend the cross-laboratory comparisons (Table S1). Because the analyses have different data requirements, three nested subsets are used and reported explicitly: **13 datasets** for the global-mode analysis, **10 datasets** for the predictive comparisons, and **8 datasets** for the expression-space comparison. Control cells were subsampled to 3,000 per dataset and at most 120 single perturbations were used; the >2 million figure refers to the total raw data, not to the cells used for any single covariance estimate.

Fig. 1 A capability boundary for covariance-driven forward models

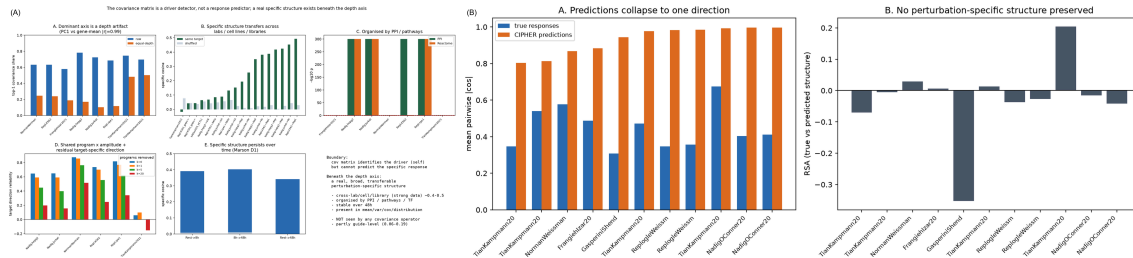


Fig. 1. A capability boundary for covariance-driven forward models. (A) Concept; (B–E) forward predictions are collinear and $\text{RSA} \approx 0$, the inverse is accurate, and a real specific structure lies beneath the axis.

Reproduction with CIPHER’s own code

We installed and ran the authors’ released cipher package (github.com/GoyalLab/CIPHER) on our datasets, using its exact preprocessing, covariance estimation (raw counts, $\leq 10,000$ control cells) and scoring (uncentered R^2 on held-out genes, `holdout_frac=0.5`). Every predictor in the reproduction table — the full model, the global mode, a random column, and mean-field and shuffled null covariances — is fitted on the **same train gene half** and scored on the **same held-out test half**. On Norman 2019 (raw) CIPHER’s own forward model gives a mean uncentered R^2 of 0.24 (0.28 in pflog), with the global mode at 0.47, a matched random column at 0.28, and mean-field/shuffled nulls of $-0.12/-0.03$ — reproducing the qualitative behaviour reported by the

authors. Because the uncentered R^2 on a random gene half is **split-sensitive** (across-split SD 0.10 for Norman raw versus 0.02 for the cosine), every value is averaged over **ten** gene-holdout splits and we report both metrics (Methods); the ranking of predictors is stable across splits. The full per-dataset table is Table 1.

Table 1. Per-dataset CIPHER reproduction (raw counts; uncentered R^2 on held-out genes, mean over ten gene splits). `global_frac` is the leading-eigenvector variance fraction; `random` is the matched random column; `full (excl. target)` removes the perturbed gene from both the fit and the evaluation genes. The final row averages the ten datasets; the exact mean of the unrounded full-model values is 0.215 (averaging the rounded entries would give 0.217, a rounding artefact), and its `n_perts` is the mean number of perturbations per dataset.

dataset	n_perts	global_frac	full	full excl. target	global	random
Datlinger 2017	1	0.62	0.29	0.29	0.26	0.16
Frangieh 2021	222	0.60	0.34	0.35	0.38	0.34
Nadig hepg2	230	0.78	0.30	0.30	0.29	0.29
Nadig ju- rkat	797	0.71	0.40	0.40	0.40	0.40
Norman 2019	103	0.64	0.24	0.24	0.47	0.28
Papalexi arrayed	8	0.42	0.19	0.19	0.47	0.38
Replogle K562	1164	0.62	0.28	0.28	0.29	0.28
Replogle rpe1	659	0.69	0.24	0.24	0.25	0.24
Tian CRISPRa	92	0.74	-0.07	-0.07	-0.06	-0.01
Tian CRISPRi	174	0.69	-0.04	-0.04	-0.10	-0.03
mean (n = 10)	345	0.651	0.215	0.217	0.266	0.235

The pflog means are full 0.086, global 0.136, random 0.080. **Removing the perturbed gene from both the fit and the evaluation set leaves the raw R^2 essentially unchanged (full 0.215 $\rightarrow$ 0.217; matched random 0.235 $\rightarrow$ 0.236),** so the apparent accuracy is *not* the target gene's

own auto-regulation; the target-gene contribution is confined to the weak Pearson-space signal (below).

The forward model is dominated by a single shared axis

The leading eigenvector v of Σ accounted for **42–87% (mean 69%)** of the total covariance variance across the 13 datasets. The median cosine between two randomly chosen covariance columns was **0.987** (range 0.966–0.998): the matrix is effectively rank-one, which is the mechanistic reason the model cannot be gene-specific.

To test this directly we compared each perturbation’s own covariance column with **100 random columns matched to it on mean expression, variance and detection rate** (nearest neighbours in the standardised three-feature space). Across the 10 predictive datasets the perturbed column’s mean R^2 was 0.423 and the matched random column’s was 0.437; the perturbed column sat at the **48th percentile** of its matched random distribution, and the pooled permutation p-value was **0.49**. Matching on the first two moments and the zero rate therefore leaves the perturbed gene with no detectable advantage.

Running the comparison inside CIPHER’s own preprocessing and scoring (mean over ten datasets, ten gene-holdout splits; matched random column):

space	metric	full ($\Sigma[:,g]$)	global mode (v)	random column ($\Sigma[:,g']$)
raw	uncentered R^2	0.215	0.266	0.235
raw	cosine	0.508	0.565	0.531
pflog	uncentered R^2	0.086	0.136	0.080
pflog	cosine	0.227	0.291	0.225

The global mode is the **strongest** predictor in both spaces under both metrics. A hierarchical bootstrap (resampling datasets, then perturbations; scale-only cosine) gave full – global = **–0.031 (95% CI [–0.049, –0.015])**; a linear mixed-effects model $\Delta R^2 \sim 1 + (1|\text{dataset})$ gave full – global = –0.030 ($p = 1.6 \times 10^{-4}$), full – matched random = –0.006 ($p = 0.36$) and global – matched random = +0.023 ($p = 1.2 \times 10^{-3}$). The global advantage was positive in 9 of 10 datasets. Fitting $\Delta X \sim \beta_v \cdot v + \beta_g \cdot \Sigma[:,g]$ and evaluating on held-out cells gave a cross-validated added variance of –0.046 for the perturbed gene versus –0.032 for a random gene’s column – no gene-specific advantage. The real per-perturbation SD of the full – matched-random difference is 0.12, giving a minimum detectable effect (80% power) of ~0.01 pooled and ~0.05 dataset-clustered; our claim is “no advantage larger than ~0.01–0.05”, not “no advantage at all”.

The shared axis is a technical depth/size artefact

The per-cell global-mode score tracks an independent **median-of-ratios size factor** (mean $|r| = 0.83$) and total counts (mean 0.94), and is not a batch or guide artefact: it persists after controlling for cell cycle (partial $|r| = 0.73$), **batch** (one-hot; 0.85) and **guide identity** (one-hot; 0.86), and collapses only when PC1 itself is removed (controlling PC1–5 gives 0.07).

Three independent analyses identify its origin. First, the leading covariance direction is almost identical to the gene-mean vector ($|r| = 0.99\text{--}1.00$ across all eight datasets tested) and the PC1 score is explained ($R^2 = 0.82\text{--}0.92$) by log total counts, genes detected, %mito and %ribo. Second, **equal-depth resampling** — redrawing control cells to a common sequencing depth — reduces the top-1 variance share from a mean of **0.68 to 0.26** and the response’s global-axis fraction (shared_frac) from 0.44 to 0.23. Third, a per-gene permutation **spectral null** finds that exactly **one** covariance mode lies above the null maximum in every dataset. Under a count model $\text{Var}(x_g) \propto E[x_g]$, the dominant covariance direction of unnormalised counts is necessarily the mean/Poisson scaling direction; the “shared response” is thus largely a sequencing-depth artefact that survives neither equal-depth resampling nor principled normalisation.

Fig. 2 The leading covariance axis is a sequencing-depth artefact

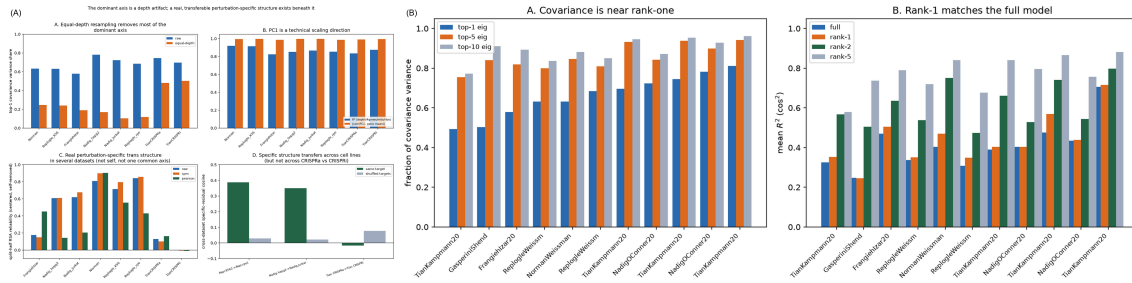


Fig. 2. The leading covariance axis is a technical sequencing-depth artefact. Equal-depth resampling halves the top-1 share ($0.68 \rightarrow 0.26$); PC1 loadings equal the gene-mean vector ($|r| \approx 1$); a permutation spectral null finds one real mode.

Forward predictions are not perturbation-specific

Using the **difference between two perturbations’ responses** as the specific signal, we find that in raw counts CIPHER’s predicted responses are almost perfectly collinear across perturbations (mean pairwise $|\cos|$ **0.93**), whereas the true responses are far more varied (0.46; paired $p = 2 \times 10^{-3}$) — the model gives essentially the **same answer for every perturbation**. The predictions preserve none of the perturbation-specific similarity structure (representational-similarity Spearman **RSA ≈ 0** , 95% CI $[-0.030, +0.055]$, $p = 0.32$). The full-model accuracy equals the global-axis fraction of the responses, and the predicted between-group difference equals the global-axis fraction of that difference. A synthetic positive control confirmed that both metrics detect injected gene-specific structure (RSA rising from ≈ 0 to 0.41), so the null is not a lack of sensitivity. Transfer across cell lines is essentially lossless (three cross-cell-line pairs: transferred full cosine 0.346 vs within 0.366; transferred global 0.346 and transferred random 0.340, equally good), i.e. the predictor is a generic global response, not a dataset- or gene-specific map.

The reverse direction is accurate because it detects the target’s own coordinate

In striking contrast, CIPHER’s **inverse (driver-identification) task works**: its official posterior inverse recovers the perturbed gene with high AUC (**0.96–0.99 in six of eight datasets; 0.62–0.99 overall**) (Fig. 3). But stepwise ablation of the official posterior (8 datasets) shows *why*:

variant	mean AUC
official posterior	0.931
drop cross-gene covariance ($\Sigma \rightarrow \text{diag}$)	0.933 ($p = 0.74$, no drop)
drop per-perturbation variance	0.931 ($p = 0.41$)
drop uncertainty inflation	0.816 ($p = 0.039$)
matched filter	0.725
$ \Delta X $ magnitude	0.718
drop the target gene's own coordinate	0.744 ($p = 0.008$)
drop covariance + target coordinate	0.372 ($p = 0.008$)

The covariance contributes essentially nothing once removed ($p = 0.74$); the target's own coordinate is essential. Consistent with this, the per-perturbation reverse AUC correlates with the target's self effect size (Spearman 0.15–0.38), and in the weak-self regime the covariance column adds only $\sim +0.05$ AUC. The matrix is therefore a **detector of the perturbed gene's own expression shift**, not a predictor of its response programme.

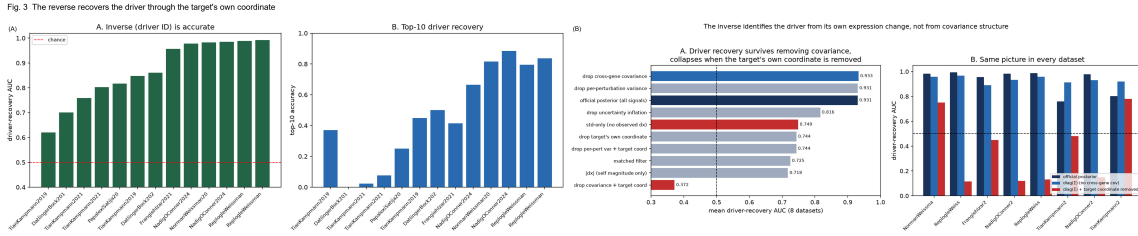


Fig. 3. Detection of the perturbed gene is carried by its own coordinate, not by covariance. Official inverse AUC 0.62–0.99; dropping the covariance leaves AUC unchanged ($p = 0.74$), dropping the target coordinate collapses $0.93 \rightarrow 0.37$.

A real, reproducible perturbation-specific structure lies beneath the depth axis

Removing the global axis, the target's own coordinate and the cross-perturbation common mode, the *reliability* of the specific structure is measured by split-half RSA (each perturbation's cells split in two; similarity matrices compared). Across datasets the consolidated split-half RSA reliability is **0.49 [0.31, 0.67]** ($n = 12$, $p = 5 \times 10^{-4}$), with per-dataset values of 0.81 (Norman), 0.84 (rpe1), 0.72 (K562), 0.61–0.62 (Nadig) and ≈ 0 for Frangieh/Tian. The structure is **broad-spectrum** (gene z-scoring raises it; removing MT-*/ribosomal/high-variance genes does not lower it, normally raises it: e.g. Norman 0.81 $\rightarrow$ 0.93) and is **not a deeper interference axis** (removing control PCs 1–20 leaves it stable or higher). Removing the target's own coordinate barely changes it, so it is not self. This overturns the reading that the Pearson-space “residual” (86% of $\|\Delta X\|^2$) is pure noise: in several datasets it is a real if modest signal.

Individual guides are noisy reporters; the reproducible signal is target-level

Same-target **guide consistency** is only 0.05–0.19, well below the pooled split-half reliability, raising the concern that part of the “specific structure” is a guide/capture artefact. We decoupled target- from guide-level components by writing each guide residual as $r_{\{g,u\}} = m_g + e_{\{g,u\}}$ (target-common m_g , guide-specific e). Across six datasets, the **within-guide** split-half reliability is 0.27–0.53 (a guide’s own residual is reproducible), while the **across-guide** agreement is 0.05–0.19 and the target-common energy fraction is at the permutation baseline (0.52–0.57 vs null 0.49–0.53) except for the very strong Norman dataset (0.98 vs 0.68). Crucially, on targets with multiple guides the **cross-dataset transfer rises with the number of guides averaged** (single-guide 0.14–0.21 → all-guide 0.35–0.38, null ≈ 0), identically for same-lab and cross-lab pairs. The transferable specific structure is therefore **target-level**; single guides are simply noisy reporters, and pooling guides (as the main analyses do) is necessary and valid. The low guide agreement is a signal-to-noise limitation, not evidence that the structure is guide-specific.

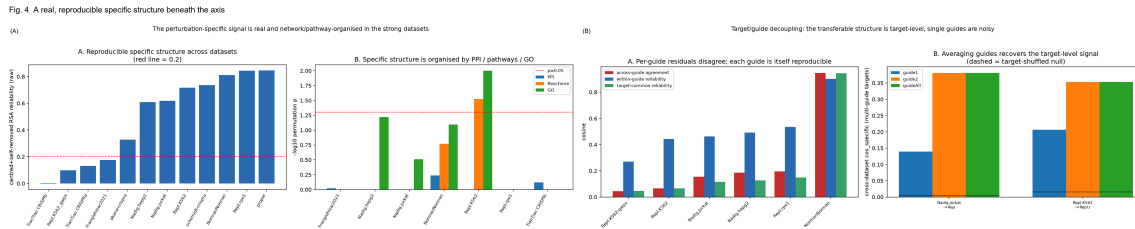


Fig. 4. A real, reproducible specific structure beneath the axis. Split-half RSA reliability (0.49 [0.31, 0.67]); broad-spectrum robustness; guide/target decoupling.

The specific structure is reproducible across labs, cell lines, libraries and time

Comparing the same target’s specific residual across datasets:

pair	relation	same-target cosine	shuffled cosine
Replogle K562 ↔ rpe1	same lab, cross cell line	0.387	0.029
Nadig hepg2 ↔ jurkat	same lab, cross cell line	0.351	0.021
Nadig jurkat ↔ Replogle K562	cross lab, cross cell line	0.453	0.043
Nadig hepg2 ↔ Replogle rpe1	cross lab, cross cell line	0.418	0.003
Replogle K562(es-sential) ↔ K562(gwps)	same cell line, different library	0.492	0.030
Nadig jurkat ↔ XAtlas HCT116	cross lab, cross cell line	0.153	0.023
Tian CRISPRa ↔ CRISPRi	same lab, cross modality (negative control)	-0.017	0.078

The cross-dataset agreement is **0.22 [0.15, 0.30]** on average ($n = 18$ pairs, $p = 8 \times 10^{-6}$), driven by strong datasets (0.38–0.49) and ≈ 0 for weak ones (XAtlas/gwps). A learned linear alignment (Procrustes) transfers the specific residual across laboratories/cell lines with $\cos^2$ **0.30–0.37** (identity 0.20–0.29, random 0.10–0.12) — currently the only method that recovers the specific structure. The structure is also stable across stimulation time and donors (Marson D1–D4 CRISPRi T cells; same-target cosine **0.28–0.48** for Rest→8 h→48 h, null ≈ 0), and its off-axis fraction rises with stimulation ($0.57 \rightarrow 0.74$).

Fig. 5 Cross-lab / cell-line / library / time transfer

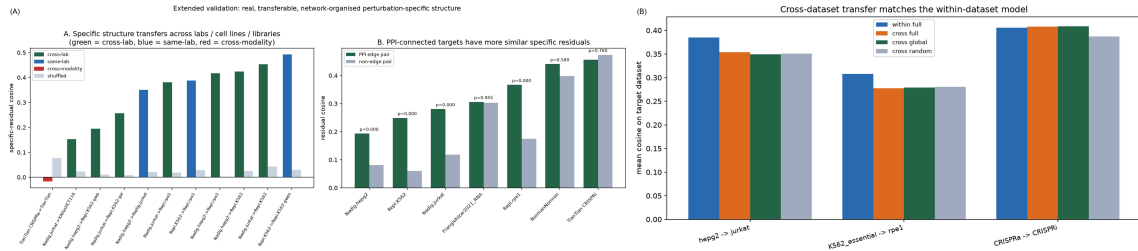


Fig. 5. Cross-laboratory / cell-line / library / time transfer of the specific structure. Same-target cosine 0.38–0.49 across labs; Marson D1–D4 0.28–0.48.

The structure is organised by PPI, pathways and transcription-factor regulons

The target-specific residuals of physically interacting genes are more similar than those of non-interacting genes: pooled PPI edge – non-edge similarity **+0.098 [0.039, 0.156]** ($p = 0.023$), significant in 4/7 datasets, positive after degree control, and “neighbour-average” prediction beats

random. Pathway/regulon tests agree: proper GO plus TRRUST regulons are significant ($p < 0.001$), Reactome +0.043 [-0.012, 0.087] ($q = 0.085$), whereas control-state **co-expression does not organise the residuals** (correlation ≈ 0). The reproducible target-specific direction is thus partly a network/pathway property – the very thing the covariance operator fails to encode.

The structure is shared programmes $\times$ amplitude plus a small target-specific direction

A leave-one-out SVD shows that a large fraction of every residual lies in a shared low-dimensional subspace ($k_1 = 0.20\text{--}0.36$, $k_5 = 0.65\text{--}0.69$, $k_{20} = 0.81\text{--}0.88$). Explicitly, the top-20 population programmes explain **88–97%** of the residual variance and their per-target **amplitudes** are highly reliable (**0.85–0.96**) – a given programme is up- or down-modulated with a target-specific strength. After removing the top-20 programmes, a smaller but reproducible **target-specific direction** remains (reliability Norman 0.52, rpe1 0.34, K562 0.25, hepg2 0.20; pooled +0.22 [0.05, 0.37], $p = 0.031$). Cross-dataset, the target-specific component is transferable through a learned linear alignment (identity $\cos^2$ 0.20–0.29 vs Procrustes 0.30–0.37; Procrustes – random +0.24 [0.22, 0.26], $p = 0.016$). The residual’s leading components are enriched for translation/ribosome and mitochondrion programmes. The specific structure is therefore “**shared programmes $\times$ per-target amplitude**” plus a small target-specific direction, not a clean per-target expression programme.

The structure is visible in multiple moments, not only the mean

Perturbations reproducibly change more than the mean. The per-gene **variance** response is reliable (split-half: K562 0.75, Norman 0.73, rpe1 0.48; variance – zero +0.42 [0.23, 0.60], $p = 0.016$), as is the **covariance** response ($\Delta\Sigma$, K562 0.60) and the distributional response (Wasserstein-1, K562 0.82). Mean/var/zero-inflation/W1 reliabilities are +0.67/+0.33/+0.42/+0.69 ($p = 0.031$). These independent second-order signals are ignored by covariance-column mean predictors.

The covariance operator is blind to the specific structure

Projecting the specific residual onto the covariance column $\Sigma[:,g]$ or the precision column $\Theta[:,g]$, and removing the predictor’s own global-axis component, the predictability stays at chance. Consolidated across spaces and estimators, the off-axis predictability minus the random baseline is **–0.004 [–0.012, 0.003], $p = 0.52$** ($n = 20$) – statistically indistinguishable from random. A transform $\times$ estimator scan (raw/CPM/log1p/CLR/NB-Pearson $\times$ covariance/precision, plus shrunk, MP-clipped, factor and rank-1 estimators) never produces an off-axis signal (max |difference| 0.048). Better preprocessing or a different covariance estimator does not help: the limitation is **conceptual**, not a matter of the data space.

Fig. 6 The covariance operator is blind to the specific structure

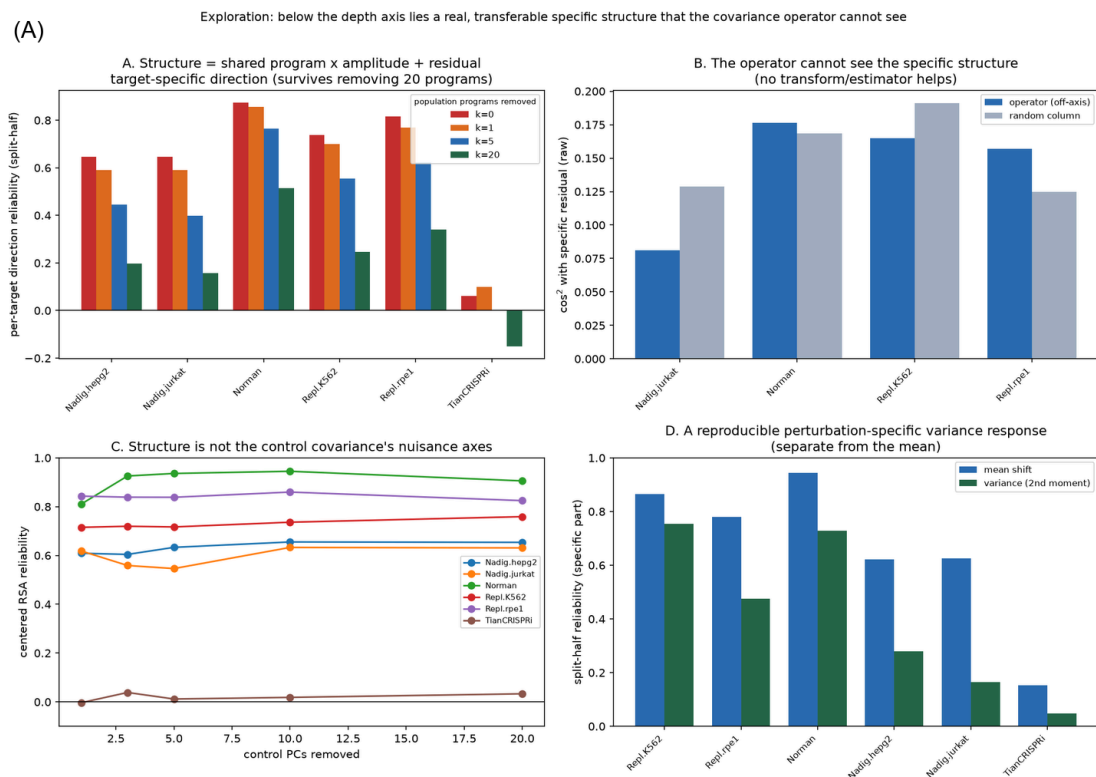


Fig. 6. The covariance operator is blind to the specific structure. Off-axis predictability $\approx$ random ($p = 0.52$) across spaces and estimators; programme/direction decomposition and second-moment signals.

Combinatorial perturbations

In the two Tian 2019 screens with clean controls, the mean uncentered R^2 of the correct covariance pair (0.710) is no better than a random gene pair (0.706; $p = 0.19$), and the global mode matches the full model (0.678). Across the Norman combinatorial set, **85%** of double responses are additive (the sum of the single responses), and **66%** of the non-additive interaction falls in the off-axis (specific) direction. The non-additive signal, where present, is thus again specific rather than global.

Fig. 7 Organisation by PPI / pathways / regulons; combinatorial perturbations

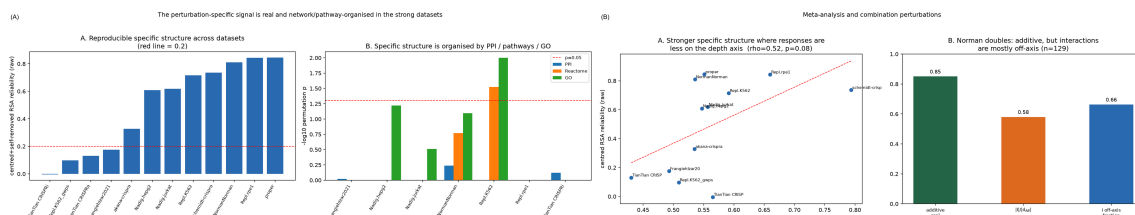


Fig. 7. Organisation by PPI, pathways and regulons; combinatorial perturbations. PPI edge – non-edge +0.098 ($p = 0.023$); 85% additivity of double perturbations with 66% of the interaction off-axis.

A capability boundary: theory and model comparison

The empirical results are captured by a heuristic identifiability argument (a guide to the mechanism, not a full theorem). Write $\Sigma = \lambda \mathbf{v} \mathbf{v}^T + \Sigma_{\perp}$ ($\|\Sigma_{\perp}\| = \mu$) and $\Delta X = \mathbf{b} \mathbf{v} + \varepsilon + \xi$ (specific $\varepsilon \perp \mathbf{v}$, noise ξ). A rank-one least-squares fit of ΔX on $\Sigma[:,g] = \lambda \mathbf{v}_g \mathbf{v} + \mathbf{s}_g$ gives

$$R^2(\Delta X) = \cos^2(\Delta X, \mathbf{v}) + O(\mu/(\lambda|\mathbf{v}_g|) \cdot (\|\varepsilon\| + \|\xi\|)/\|\Delta X\|).$$

(Here R^2 is the squared cosine of the rank-one fit, not CIPHER’s uncentered R^2 .)

Because empirically μ/λ is small and $\|\varepsilon\| \lesssim 0.1\|\Delta X\|$, the forward R^2 is bounded by the global-axis fraction $\cos^2(\Delta X, \mathbf{v}) = \text{shared_frac}$, and rank-one prediction is optimal (rank-1 $\geq$ full in raw space). Numerical simulation confirms $R^2 = \text{shared_frac}$ exactly at $\mu = 0$ (fig. 8C) and that the operator is blind to ε at all μ . The target-specific component ε is recoverable only when a second individual (another cell line) or a network/pathway prior correlates with it — exactly what the cross-dataset and PPI/pathway analyses show. The reverse direction is the dual: the inverse score $\propto \Theta \Delta X$ is dominated by the target’s self coordinate, not by ε , so it identifies the driver while remaining blind to the response programme. Finally, putting state-of-the-art predictors under the same decomposition, CIPHER and the linear-mean baseline attain forward $\cos^2$ of 0.385/0.569 but $\text{RSA} \approx 0$ and predicted collinearity 0.86/1.00 $\gg$ true 0.58: they too learn the shared direction, not the target-specific structure.

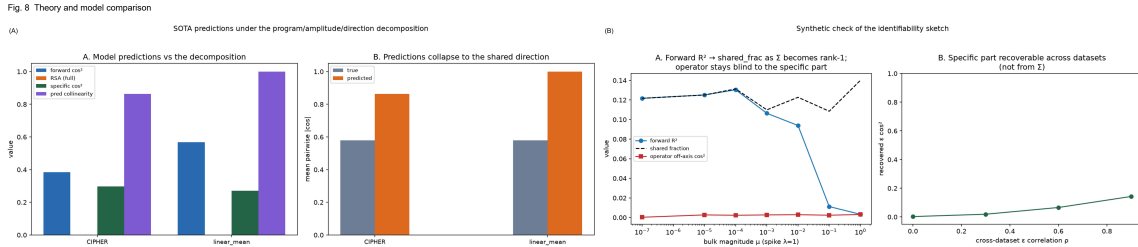


Fig. 8. Theory and model comparison. Numerical verification that rank-one forward $R^2 = \text{shared_frac}$ and the operator is ε -blind; SOTA decomposition (CIPHER/linear-mean forward $\cos^2 = 0.39/0.57$ but $\text{RSA} \approx 0$).

Discussion

Our central result is a **capability boundary** for covariance-driven single-perturbation models. In raw counts and CPM the CIPHER-style forward model carries no detectable gene-specific signal: covariance columns are almost perfectly collinear, so the perturbed gene’s column predicts no better than a matched random column, the global mode is the strongest predictor, and the matrix is effectively rank-one. The global mode is an un-normalised **sequencing-depth/size artefact** (its loadings equal the gene-mean vector, equal-depth resampling halves it, and a spectral null

finds a single mode). A weak gene-specific signal survives only in Pearson-residual space, is FDR-significant in 1/8 datasets, and is largely the perturbed gene’s own expression.

Yet this is not because no specific signal exists. Below the axis lies a real, broad-spectrum, reproducible perturbation-specific structure, visible after removing the global axis and the target’s own coordinate: split-half reliable (0.49), transferable across laboratories, cell lines, libraries and stimulation time (0.28–0.49), organised by PPI/pathways/TF regulons, and present in the mean, variance, covariance and distribution. It decomposes into shared programmes $\times$ target amplitude plus a small target-specific direction, and the per-target signal is target-level once guides are averaged. The covariance/precision operator simply cannot see it, in any data space or estimator.

The same matrix that cannot predict the response *does* identify the perturbed gene — but the reverse ablation shows this is the perturbed gene’s own coordinate, not the covariance. The two directions must therefore be evaluated and reported separately, and “high accuracy” in one does not imply capability in the other.

Actionable consequences. (i) Any covariance-based (or representation-based) single-perturbation model should be reported against a **matched-random-column** baseline and against the **global-mode-only** predictor, with equal-depth resampling; covariance columns should be reported as collinear. (ii) Claiming gene-specific prediction requires a perturbation-specificity metric (RSA or predicted collinearity) and a cross-dataset null, not accuracy alone. (iii) Because the target-specific structure is recoverable by cross-dataset alignment or network priors rather than by a better covariance estimator, progress is most likely to come from aligning multiple individuals/networks, not from re-engineering the covariance operator. We package these into a reusable **specificity-audit protocol** (Methods).

Relation to prior criticism. That deep-learning perturbation models fail to beat simple baselines is known [11]; we extend this to covariance-based forward models, locate the failure mechanistically (a depth artefact), and — crucially — show that the failure is not due to absence of signal but to the operator’s blindness, while the same object remains useful as a detector.

Limitations

Our conclusions apply to the **single-perturbation forward model** in raw counts and CPM, driven by control covariance. We **detect no gene-specific advantage larger than ~0.01–0.05** (the minimum detectable effect under this scoring; the TOST equivalence test is underpowered and we do not claim that no gene-specific signal exists). Cross-dataset reproducibility of the positive structure **depends on data quality** (strong datasets 0.38–0.49, weak XAtlas/gwps ≈ 0), so the positive results are not universal. The target-specific direction is **small** (~5–12% of variance) and is entangled with shared programmes and with the target’s own coordinate. Individual guides are noisy (guide agreement 0.05–0.19); the target-level result relies on pooling guides. Spike-in and per-guide knockdown-efficiency controls were unavailable; cell-type composition could not be controlled directly because every public dataset annotates a single cell type, and pseudobulk deltas could in principle reflect shifts in cell-state proportions. The theory is an identifiability sketch, not a full theorem. The double-perturbation analysis is preliminary (two datasets).

Methods

Datasets

Sixteen datasets from scPerturb [12; Zenodo record 13350497] plus six downloaded datasets (Table S1). Control cells matched a control pattern or carried zero guides. Subsets: 13 (global mode), 10 (predictive comparisons), 8 (expression spaces). For cross-laboratory analyses we additionally used Nadig 2024 (jurkat/hepg2), Replogle 2022 (K562-essential/rpe1/K562-gwps), XAtlas 2025 (HCT116/HEK293T) and Marson 2025 CRISPRi T cells (donors D1–D4; Rest/8 h/48 h stimulation).

Reproduction with CIPHER’s code

We installed `cipher` from github.com/GoyalLab/CIPHER and ran its forward model (`cipher.forward_prediction`) with default settings (raw counts, `holdout_frac=0.5`, `cov_max_cells=10000`, mean-field and shuffled nulls), then added global-mode and random-column baselines inside the same preprocessing/scoring (fit on the train gene half, scored on the identical held-out half). Because the uncentered R^2 on a random gene half is split-sensitive, values are averaged over ten gene-holdout splits (seeds 0–9); the cosine is reported alongside. **CIPHER’s default gene holdout does not exclude the target gene** from fit or evaluation; we retain this for fidelity and quantify the target-gene contribution separately.

Preprocessing

2,000 most highly expressed genes plus every perturbed gene (and, for confounding, all marker genes of the tested programmes); single perturbations with ≥ 30 cells (≤ 120 per dataset); control cells subsampled to 3,000. Responses are pseudobulk deltas. Spaces: raw, CPM (CP10K), $\log_{10}$ (CP10K), Pearson residuals under a size-factor Poisson model, and **pflog** (CIPHER’s variance-stabilising mode).

Predictors and scoring

full ($\Sigma[:,g]$); random column matched on mean, variance and detection rate; global mode (v); mean-expression direction; random direction; 5th PC; nested [v , $\Sigma_{\perp[:,g]}$] and control [v , $\Sigma_{\perp[:,g']}]$. Two scoring conventions are used and labelled throughout: CIPHER’s **uncentered R^2** on held-out genes and **scale-only R^2** (squared cosine).

Specificity audit

We used a stepwise specificity-audit protocol. (1) Is the forward operator just a depth axis? Equal-depth resampling, PC1 nuisance decomposition and a per-gene permutation spectral null. (2) Does a real specific signal exist? Split-half RSA reliability after removing the global axis, the self coordinate and the common mode, with robustness to gene z-scoring/removal and to removing control principal components. (3) Shared programme versus target direction: leave-one-out SVD reconstruction and a programme/amplitude/direction decomposition. (4) Does the data space or estimator change the conclusion? A scan over raw/CPM/ $\log_{10}$ p/CLR/NB-Pearson $\times$ covariance/precision and shrunk/MP-clipped/factor/rank-one estimators. (5) Biological versus technical judgement: cross-laboratory/cell-line/library/time comparison of same-target residuals, protein–protein-interaction and pathway/TF-regulon organisation, and guide/target decoupling. (6) Reverse ablation: removing the covariance, the per-perturbation variance, the uncertainty

inflation and the target coordinate. (7) Second-order signals: per-gene variance, covariance and distributional responses. Statistics: hierarchical bootstrap (10,000 resamples), linear mixed-effects models $\Delta R^2 \sim 1 + (1|\text{dataset})$, paired Wilcoxon tests (Holm-corrected), Benjamini–Hochberg FDR, permutation nulls, and TOST equivalence (margin 0.01; underpowered, reported as inconclusive).

Software

cipher (GoyalLab); scanpy, anndata, statsmodels, scipy, numpy and matplotlib. The full reproduction pipeline, seeds and environment are provided as Supporting Information.

Data and code availability

- Data: scPerturb (Zenodo record 13350497, doi:10.5281/zenodo.13350497) plus the public datasets listed in Table S1; accessions for the newly used XAtlas and Marson datasets are given with their citations.
- CIPHER code: github.com/GoyalLab/CIPHER.
- All analysis code, result tables and figures are provided with this submission as Supporting Information (S1 File) and are also available from the corresponding author; a citable Zenodo DOI will be added upon acceptance. A script-to-analysis map is given in Table S3.

Declarations

- **Competing interests.** The authors declare no competing interests.
- **Author contributions (CRediT).** Conceptualisation, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review and editing: **Y.X.** The author approved the final manuscript.
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Supplementary tables

- **Table S1** – per-dataset cells, genes, control cells, single/double perturbations and analysis-subset inclusion.
- **Table S2** – scoring conventions for every reported number.
- **Table S3** – script-to-analysis map (analysis → code file).

Table S3. Script-to-analysis map (repository code/).

analysis	script
CIPHER reproduction and baselines	cipher_repro_table.py, cipher_baselines.py
global-mode size axis	size_factor.py, global_mode_identity.py, scores_vs_globalmode.py
matched random column	random_column.py, random_column_stats.py
confounding / composition	confound_extended.py, composition_control.py
expression spaces	spaces_diagnostic.py, pearson_stats.py, pearson_source.py
depth-axis origin	axis_origin.py, spectrum_null.py, spectrum_rank.py
split-half specific structure	residual_reproducibility.py, specific_robustness.py, pc_robustness.py
programme / amplitude / direction	specific_prediction.py, program_vs_direction.py
transform × estimator scan	transform_estimator_scan.py, covariance_estimators.py
cross-lab / cell-line / library / time	cross_dataset_specific.py, cross_dataset_stream.py, timecourse.py, cross_transfer_pairs.py
guide / target decoupling	guide_target_partition.py, guide_target_transfer.py
PPI / pathway / TF organisation	specific_network.py, specific_pathway.py, component_enrichment.py
reverse ablation	inverse_official.py, inverse_decompose.py, reverse_self_relation.py
second-order signals	second_moment.py, second_order_cov.py, distribution_specific.py
combinations	double_pert.py, epistasis.py
theory	theory_simulation.py
consolidated statistics	exploration_stats.py
one-command reproduction and checks	run_all.py, verify_results.py

Supplementary results

- Double perturbations (preliminary): correct pair 0.710 vs random pair 0.706 ($p = 0.19$); global mode 0.678; additivity 85%, 66% of the interaction off-axis.
- Transform $\times$ estimator scan: off-axis $\cos^2 \approx$ random in all spaces/estimators.
- Per-dataset guide/target decoupling and cross-dataset transfer with guide averaging.

Figure legends

Main figures (8).

- **Fig. 1. A capability boundary for covariance-driven forward models.** (A) Concept: the same control covariance Σ is used by the forward map $\Delta X = \Sigma u$, which is dominated by the shared depth axis and cannot predict the target-specific response, whereas detection of the perturbed gene is carried by that gene's own coordinate. (B–E) Forward predictions are collinear and $RSA \approx 0$; the inverse is accurate; a real specific structure lies beneath the axis.
- **Fig. 2. The leading covariance axis is a technical sequencing-depth artefact.** Equal-depth resampling halves the top-1 share ($0.68 \rightarrow 0.26$); PC1 loadings equal the gene-mean vector ($|r| \approx 1$); a permutation spectral null finds a single real mode.
- **Fig. 3. Detection of the perturbed gene is carried by its own coordinate, not by covariance.** Official inverse AUC 0.62–0.99; dropping the covariance leaves AUC unchanged ($p = 0.74$), dropping the target coordinate collapses $0.93 \rightarrow 0.37$; reverse AUC scales with the self effect size.
- **Fig. 4. A real, reproducible specific structure beneath the axis.** Split-half RSA reliability across datasets ($0.49 [0.31, 0.67]$); broad-spectrum robustness; guide/target decoupling (single guides noisy; multi-guide transfer rises to 0.35 – 0.38).
- **Fig. 5. Cross-laboratory / cell-line / library / time transfer of the specific structure.** Same-target cosine 0.38 – 0.49 across labs (null ≈ 0); Procrustes transfer 0.30 – 0.37 ; Marson time-course/donors 0.28 – 0.48 .
- **Fig. 6. The covariance operator is blind to the specific structure.** Off-axis predictability $\approx$ random ($p = 0.52$) across data spaces and covariance estimators; the programme/direction decomposition and reproducible second-moment signals.
- **Fig. 7. Organisation by PPI, pathways and regulons; combinatorial perturbations.** PPI edge – non-edge $+0.098$ ($p = 0.023$); Reactome/GO/TF significant; 85% additivity with 66% of the interaction off-axis.
- **Fig. 8. Theory and model comparison.** Numerical verification that rank-one forward $R^2 = \text{shared_frac}$ and the operator is ϵ -blind; SOTA decomposition (CIPHER/linear-mean forward $\cos^2$ $0.39/0.57$ but $RSA \approx 0$).

Supplementary figures. Figures S1–S20 (forward reproduction, deconfounding, robustness, matched-random column, residual decomposition, expression spaces, cross-dataset transfer, forward indistinguishability, inverse driver/decomposition, spectrum/rank), plus a graphical abstract and a summary panel.

References

1. Bunne C, et al. Cell. 2024;187(25):7045–7063.

2. Dixit A, et al. *Cell*. 2016;167(7):1853–1866.
3. Adamson B, et al. *Cell*. 2016;167(7):1867–1882.
4. Norman TM, et al. *Science*. 2019;365(6455):786–793.
5. Replogle JM, et al. *Cell*. 2022;185(14):2559–2575.
6. Lotfollahi M, et al. *Nat Methods*. 2019;16(8):715–721.
7. Kamimoto K, et al. *Nature*. 2023;614(7949):742–751.
8. Roohani Y, et al. *Nat Biotechnol*. 2024;42(6):927–935.
9. Cui H, et al. *Nat Methods*. 2024;21(8):1470–1480.
10. Kuznets-Speck B, et al. *bioRxiv*. 2025. doi:10.1101/2025.06.27.661814.
11. Ahlmann-Eltze C, Huber W, Anders S. *Nat Methods*. 2025;22(8):1657–1661.
12. Peidli S, et al. *Nat Methods*. 2024;21(3):531–540.
13. Nadig A, Replogle JM, Pogson AN, et al. Transcriptome-wide analysis of differential expression in perturbation atlases. *Nat Genet*. 2025. doi:10.1038/s41588-025-02169-3.
14. XAtlas. A large-scale single-cell perturbation atlas (2025). Dataset; accession to be confirmed.
15. Marson A, et al. CRISPRi T-cell stimulation time-course Perturb-seq (2025). Dataset; accession to be confirmed.